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Sacral Nerve Stimulation and Chronic Bladder Pain: Meta-Analysis

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ABSTRACT

Objectives: Chronic bladder pain syndrome (CBPS) is a debilitating condition with limited treatment efficacy. This meta-analysis evaluates the effectiveness of sacral nerve stimulation (SNS) in the management of pain, urinary symptoms, and quality of life (QoL) in patients with CBPS.

Materials and Methods: A systematic review was conducted across EMBASE, PubMed, and the Cochrane Library, adhering to Preferred Reporting Items for Systematic reviews and Meta-Analyses guidelines. Studies involving adult patients with bladder pain symptoms treated with SNS and reported pain scores were included. Pain scores were the primary outcome measure of interest. Secondary outcome measures included Interstitial Cystitis Problem Index (ICPI)/Interstitial Cystitis Symptom Index (ICSI), urinary frequency, nocturia, and QoL. Risk of bias was assessed using Risk Of Bias In Non-randomized Studies—of Interventions, and pooled effect estimates were calculated using random-effects meta-analysis models.

Results: A total of 15 studies ($N = 494$) were included in the meta-analysis. SNS was associated with a statistically significant reduction in pain scores (mean difference: -2.27 on a 0–10 scale, 95% CI: -2.94 to -1.60 , $p < 0.001$), ICSI (-2.18 , 95% CI: -2.99 to -1.37 , $p < 0.001$), urinary frequency (-1.71 voids/d, 95% CI: -2.29 to -1.12), and nocturia (-1.49 episodes/night, 95% CI: -2.35 to -0.63). Changes in ICPI and QoL were not statistically significant. Complication rates ranged from 0% to 40.6%, and explantation rates from 0% to 18%.

Conclusions: This meta-analysis suggests that SNS may offer clinically meaningful pain and urinary symptom relief in patients with CBPS. However, substantial methodological heterogeneity, moderate-to-high risk of bias, and the absence of randomized control trials limit the strength of conclusions. Larger, high-quality trials with standardized diagnostic criteria and outcome measures are essential to establish SNS as a validated treatment option for CBPS.

Keywords: Chronic bladder pain syndrome, chronic pain, neuromodulation, sacral nerve stimulation

INTRODUCTION

Chronic bladder pain syndrome (CBPS) is a debilitating condition that affects approximately 6% of the global population, with a fivefold higher prevalence in women.¹ CBPS is characterized by persistent or recurrent pain perceived in the urinary bladder region but is typically accompanied by other symptoms, such as pain that worsens with bladder filling, and urinary urge and/or urgency with associated frequency during the day and night. Unlike most other bladder conditions, there is no proved infection or other pathologic cause for these symptoms.²

Despite the high prevalence, many cases go undiagnosed or are misattributed to conditions such as recurrent urinary tract infections or overactive bladder (OAB), complicating effective management.³ Patients often experience symptoms for years before receiving a correct diagnosis, underscoring the urgent need for better diagnostic tools and treatment strategies.

Previously known as interstitial cystitis (IC), CBPS remains challenging to treat because the condition encompasses a heterogeneous spectrum of disorders, which complicates both diagnosis and treatment. Its complex and multifactorial nature requires a comprehensive approach that addresses both pain and

bladder dysfunction for effective management. CBPS also is linked to a wide range of adverse effects, including cognitive, behavioral, sexual, and emotional impacts.⁴

The treatment modalities for CBPS are varied, reflecting the complexity of the condition.⁵ No single treatment has proved universally effective. Management often involves a multimodal and multidisciplinary approach that includes physical interventions, pharmacotherapy, lifestyle changes, and psychologic interventions. Treatment aims to manage symptoms rather than offer a primary

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Table 1. Inclusion and Exclusion Criteria.

Inclusion criteria	Exclusion criteria
Studies evaluating the use of SNS in patients with chronic bladder pain symptoms (CBPS did not have to have been formally diagnosed).	Nonoriginal articles.
RCTs.	Duplicate reports.
Comparative, retrospective, and prospective studies.	Case reports.
SNS vs sham/placebo/other intervention	Animal models.
Use of pain scoring system or indicator.	<3 mo follow-up.
>3 mo of pain symptoms.	Articles without full text.
Age >18 y.	

cure. Unfortunately, many patients undergo multiple treatment approaches with limited or no relief, often progressing to more invasive interventions that also offer no guaranteed success.⁶

One of the less invasive techniques for managing pain in CBPS is sacral nerve stimulation (SNS). SNS is a neuromodulation technique that has been used to treat bladder disorders by applying electrical stimulation to the sacral nerves. SNS has been used successfully in the management of bladder dysfunctions, such as OAB and urinary incontinence. It is believed to act by modulating the neural pathways involved in bladder control and reducing both the frequency and urgency.^{7,8} The actual mechanism through which neuromodulation alleviates chronic pain is not fully understood. However, it has been shown to influence both peripheral and central neural pathways involved in pain transmission and sensory function.⁹ Other forms of neuromodulation include spinal cord stimulation (SCS), which has proved to be successful in conditions such as persistent spinal pain and chronic regional pain syndromes.^{10,11} Consequently, in the past two decades, it was hypothesized that by stimulating the sacral nerves, SNS may help to modulate abnormal neural activity that contributes to both bladder dysfunction and pain.

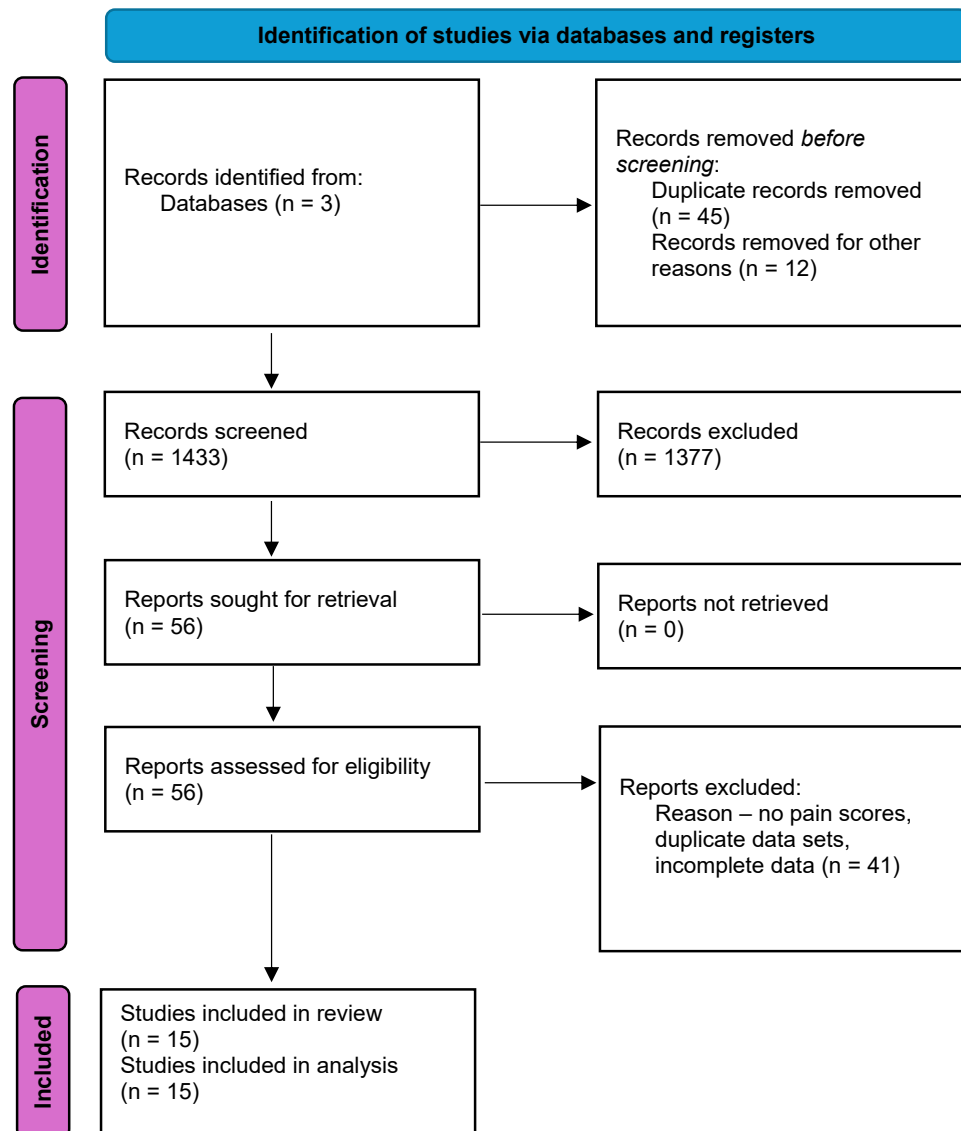


Figure 1. Flow sheet of study retrieval. [Color figure can be viewed at www.neuromodulationjournal.org]

Table 2. Primary and Secondary Outcome Measures.

Primary outcome measure	Secondary outcome measure
Pain score	QoL indicators ICPI/ICSI Urinary frequency Nocturia

The European Association of Urologists (EAU) suggests SNS may have a role in managing patients with urgency urinary incontinence refractory to antimuscarinic therapy, and there is evidence that SNS is able not only to improve bladder control but also to reduce bladder pain in some of these patients.² Some studies have shown SNS to significantly improve quality of life (QoL), urinary frequency, and pain scores in patients with bladder dysfunction and pain.¹² Moreover, a broader review of chronic pelvic pain (CPP) syndromes found that SNS was generally effective in reducing pain score. However, the authors found treatment resistance and less pain alleviation in patients with CBPS.¹³ This variability in patient response further complicates our understanding of the effectiveness of SNS in treating CBPS. Although the existing evidence is promising, SNS remains an “off-label” treatment for CBPS despite its growing use.

Considering the significant impact of chronic bladder pain on patients’ QoL and the limitations of existing evidence, a more thorough evaluation of SNS is warranted. Recent advancements in neuromodulation technology such as optimized neural dosing, high-frequency stimulation, and closed loop stimulation over the past decade further justify a reassessment of its therapeutic potential. In this study, we conducted a systematic review of published literature and performed a meta-analysis to evaluate the effectiveness of SNS in reducing pain, improving QoL, and alleviating urinary symptoms in patients with CBPS. This analysis seeks to offer a more comprehensive assessment of SNS’s therapeutic potential and its limitations, providing clearer insight for clinical practice and identifying gaps for future research.

MATERIALS AND METHODS

A systematic search was conducted across EMBASE, PubMed, and the Cochrane Library following Preferred Reporting Items for Systematic reviews and Meta-Analyses guidelines. The search was conducted by two authors, with no date exclusionary points. Search terms were developed around two key concepts: CBPS and SNS. The search was initially conducted in PubMed and later adapted for use in the other data bases.

Our inclusion criteria focused on all studies evaluating SNS in adult patients with chronic bladder pain symptoms; CBPS did not have to have been formally diagnosed. This included randomized controlled trials (RCTs), prospective studies, retrospective studies, and case series, provided that these studies used a pain scoring system. Exclusion criteria encompassed case reports, animal studies, articles with follow-up periods of <three months, and studies with incomplete or inaccessible text (Table 1).

Data records were exported to an EndNote library for initial deduplication. Two independent reviewers screened the titles and abstracts using a predefined set of questions to assess the relevance of each study (Supplementary Data Appendix S1). The references of studies also were reviewed to identify any additional relevant articles that may not have been captured by the initial data base search. We also considered relevant conference proceedings and abstracts. Interrater reliability was ensured by requiring both reviewers to pass a consistency check before proceeding with full-scale screening. In cases of disagreement during either the title/abstract or full-text screening stages, consensus was reached through discussion or, if necessary, adjudicated by a third reviewer. The risk of bias for RCTs was assessed using the Cochrane Risk of Bias 2 tool, whereas nonrandomized studies were evaluated with the Risk Of Bias In Nonrandomized Studies—of Interventions instrument.

A standardized data extraction form was developed by the review team. Each included study was reviewed independently by ≥two reviewers, who extracted data on trial design, sample characteristics, intervention specifics, and both primary and secondary outcomes. The extraction process also gathered information on blinding, randomization methods, control group types, and the handling of

Table 3. Summary of Study Characteristics (1).

Authors	Year	Study type	Total participants	Sample size*	Age, y	Sex (% female)	Symptom (mo)
Liu et al ¹⁶	2022	Prospective cohort	40	15	53	95	NA
Hernandez et al ¹⁷	2021	Retrospective observational	19	12	52.9	76	NA
Meng et al ¹⁴	2020	Retrospective cohort	203	203	52.2	54.2	NA
Marinkovic ¹⁸	2019	Retrospective cohort	65	48	48	100	NA
Zhang et al ¹⁹	2018	Retrospective cohort	81	59	55.7	81.4	60
Ghazwani ²⁰	2011	Retrospective cohort	21	11	44	100	36
Marinkovic ¹⁸	2011	Retrospective cohort	34	30	41	100	93
Peters et al ¹⁵	2008	Prospective cohort	93	20	52	90	NA
Chung et al ²¹	2007	Prospective cohort	30	17	58.5	47	6
Kessler et al ²²	2007	Prospective cohort	7	7	57	93	NA
Comiter ²³	2003	Prospective cohort	25	17	47	94	NA
Whitmore ²⁴	2003	Prospective cohort	33	17	44	NA	49
Maher et al ²⁵	2001	Prospective pilot case series	15	15	62	100	NA
Feler ²⁶	1999	Prospective case series	17	10	NA	NA	150
Shaker ²⁷	1998	Prospective case series	18	18	42.3	89	79

NA, not available.

*The total number of participants who underwent full implant with complete follow-up.

Table 4. Summary of Study Characteristics (2).

Author	Primary diagnosis	Complication rate %	Explantation %	Intervention level	Frequency (Hz)	Follow-up time (mo)
Liu et al ¹⁶	IC	0	0	S3/S4	NA	12
Hernandez et al ¹⁷	IC/CBPS	NA	NA	S3	NA	75
Meng et al ¹⁴	Pelvic floor dysfunction	19.2	NA	NR	18.4	21.0
Marinkovic ¹⁸	CPP	9.2	NA	S3	NA	116
Zhang et al ¹⁹	IC/CBPS	23.7	NA	S3	NA	20.1
Ghazwani ²⁰	CPBS	18	0	S3	14	71
Marinkovic ¹⁸	IC	27	NA	S3	NA	86
Peters et al ¹⁵	OB, CPP	NA	NA	NR	NA	6
Chung et al ²¹	OAB, IC, CPP	35	17.6	S3	14	21.6
Kessler et al ²²	CPP	NA	NA	S4	NA	24
Comiter ²³	IC	0	0	S3	16	14
Whitmore ²⁴	IC	NA	NA	s3	NA	0
Maher et al ²⁵	IC	NA	0	S3	15	62.4
Feler ²⁶	IC	11.7	5.8	S2/S3/S4	NA	11.7
Shaker ²⁷	OAB	0	5.5	S3	15	18

NA, not available; NR, not reported.

missing data. Any disagreements during data extraction were resolved through discussion between the reviewers (Fig. 1).

The primary outcome measure centered on pain relief, assessed through scoring systems such as the numerical rating scale (NRS). Secondary outcomes included evaluations of QoL, urinary frequency, nocturia, and the Interstitial Cystitis Problem Index/Interstitial Cystitis Symptom Index (ICPI/ICSI) (Table 2).

Statistical Analysis

We conducted meta-analyses to synthesize findings across studies and compute pooled effect estimates of the association between sacral neuromodulation and the primary and secondary outcomes. To ensure comparability across studies, visual analog scale and QoL scores were transformed to a scale of 0 to 10 for calculation of pooled mean differences. In sensitivity analysis, we used standardized mean differences (SMDs) as the summary effect measure. SMDs and corresponding 95% CIs were calculated for each study on the basis of reported means, SDs, and sample sizes. For the two studies that reported effect sizes by subgroups, a weighted average of the subgroup effect sizes was calculated and included in the meta-analysis. One study¹⁴ reported results separately for men and women, and the other study¹⁵ reported results for responders to sacral neuromodulation and nonresponders.

Statistical heterogeneity among studies was assessed using the Cochran's Q test (with a significance threshold of $p < 0.10$) and quantified using the I^2 statistic. All meta-analyses showed high heterogeneity ($I^2 > 70\%$), suggesting substantial variability in effect sizes across studies. Random-effects models based on the DerSimonian and Laird method were therefore used for all

meta-analyses to account for between-study variability, which was anticipated given methodological and clinical differences among studies. Most of the included studies had small sample sizes, which may have reduced the precision of individual effect estimates and contributed to the observed heterogeneity. This limitation was considered during the interpretation of results. Forest plots were generated to visualize individual study estimates and pooled results. All meta-analyses were performed in Stata version 17 (StataCorp, College Station, TX) using the metan command. Two-tailed p -values < 0.05 were considered statistically significant.

RESULTS

Overall Efficacy

The meta-analysis synthesized data from 15 studies evaluating the impact of SNS in CBPS. SNS was associated with significant improvements in pain scores and urinary symptoms (Tables 3, 4, and 5). Pain relief showed an average reduction of 2.27 points (95% CI: $-2.94, -1.60$) on a scale of 0 to 10. Despite these improvements, complication rates varied widely, ranging from 0% to 40%, and the explantation rate was reported as high as 18%. In addition, the heterogeneity in study design and methodological variability was a limiting factor.

Diagnosis and Demographics

A cumulative population of 494 participants was included, with sample sizes ranging from seven to 203 patients. All participants showed some evidence of CBPS with/without dysfunction. However, diagnostic criteria varied, including overlapping conditions

Table 5. Outcome Measure Summary.

Outcome	Pooled effect estimate	Scale range standardized	Clinical interpretation
Pain	-2.27 (95% CI: -2.94 to -1.60 $p < 0.05$)	NRS scale 0–10	Statistically significant improvement in pain score
ICPI	-1.15 (95% CI: -3.02 to 0.72 $p = 0.23$)	ICPI scale 0–20	Not statistically significant improvement in symptoms
ICSI	-2.18 (95% CI: -2.99 to -1.37 $p < 0.05$)	ICSI scale 0–20	Statistically significant improvement in symptoms
QoL	-1.46 (95% CI: -4.27 to 1.35 $p = 0.31$)	QoL 0–100	Not statistically significant worsening in QoL
Voiding frequency	-1.71 (95% CI: -2.29 to -1.12 $p < 0.05$)	Voids/24 h	Statistically significant voiding reduction
Nocturia	-1.49 (95% CI: -2.35 to -0.63 $p < 0.05$)	Episodes per night	Statistically significant improvement in nocturia

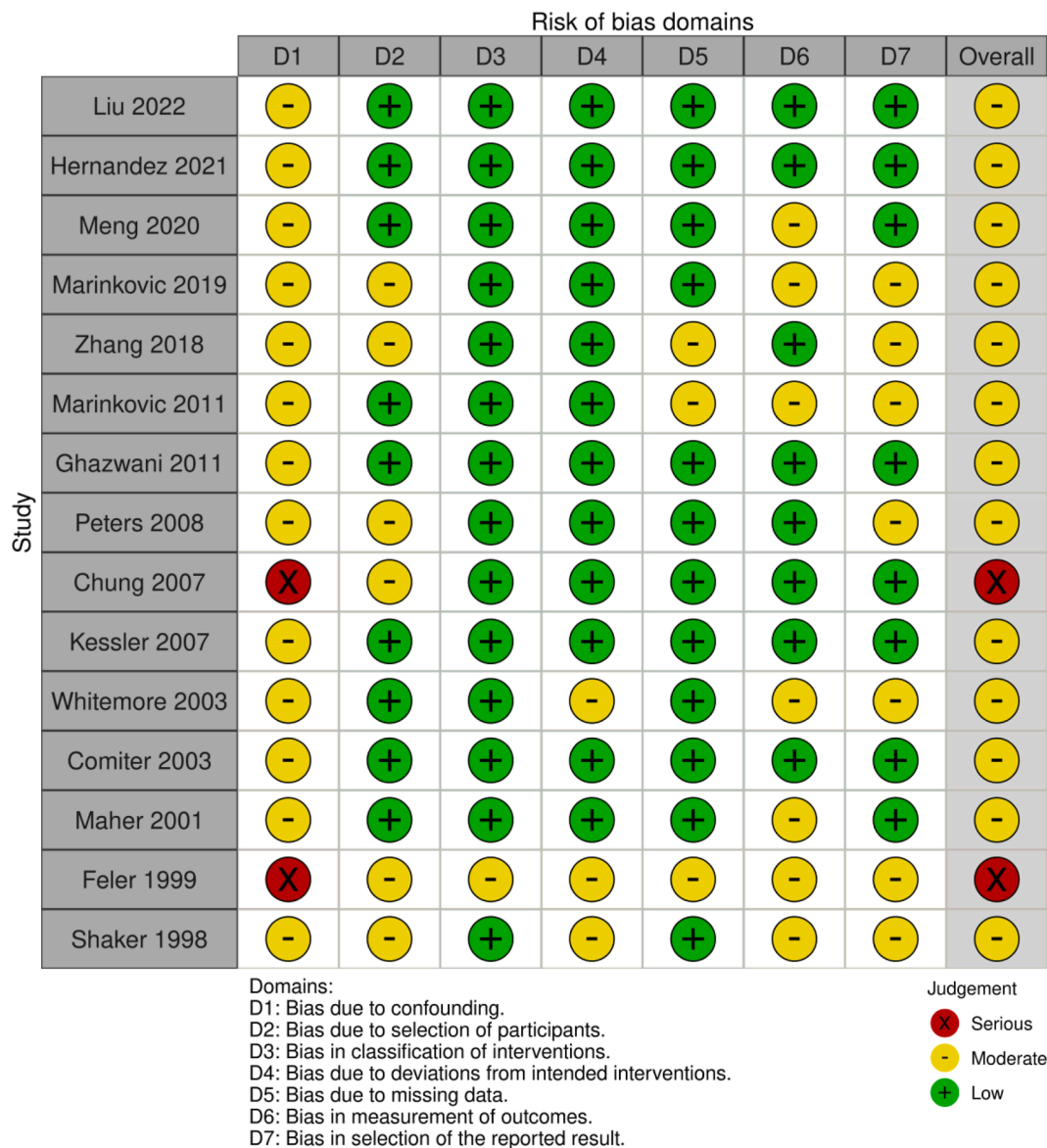


Figure 2. Risk of bias traffic light plot. [Color figure can be viewed at www.neuromodulationjournal.org]

such as CPP and OAB. The diagnostic criteria for CBPS varied significantly, with some studies failing to use standardized definitions or explicitly report diagnostic methods.

- Age: The mean participant age across studies ranged from 42.3 to 62 years, with an overall weighted mean of 51.7 years.
- Sex: Women represented 53% to 100% of the study populations, reflecting the higher prevalence of CBPS in female patients.
- Symptom duration: Baseline symptom duration ranged from six to 150 months.

Study Designs and Statistical Considerations

The analysis comprised nine prospective cohort studies/case series and six retrospective cohort studies. No RCTs were available, and only one study implemented partial randomization without blinding. Statistical heterogeneity was substantial ($I^2 = 78\%$,

$p < 0.01$), driven by variations in diagnostic criteria, intervention protocols, and outcome measures.

- Follow-up durations: Follow-up periods ranged from six months to 116 months, with a median follow-up of 24 months.
- Risk of bias: Most studies were classified as having moderate-to-high risk of bias owing to small sample sizes, lack of blinding, and incomplete reporting of outcomes (Fig. 2).

Pain Relief (NRS)

The mean baseline NRS score was 7.6, which decreased to a postintervention score of 3.3, reflecting a calculated SMD decrease of -2.27 points (95% CI: $-2.94, -1.60$, $p < 0.005$) (Figs. 3 and 4).

- Largest reductions: Kessler et al²² reported an NRS SMD reduction of -5.62 (95% CI: $-7.92, -3.32$).

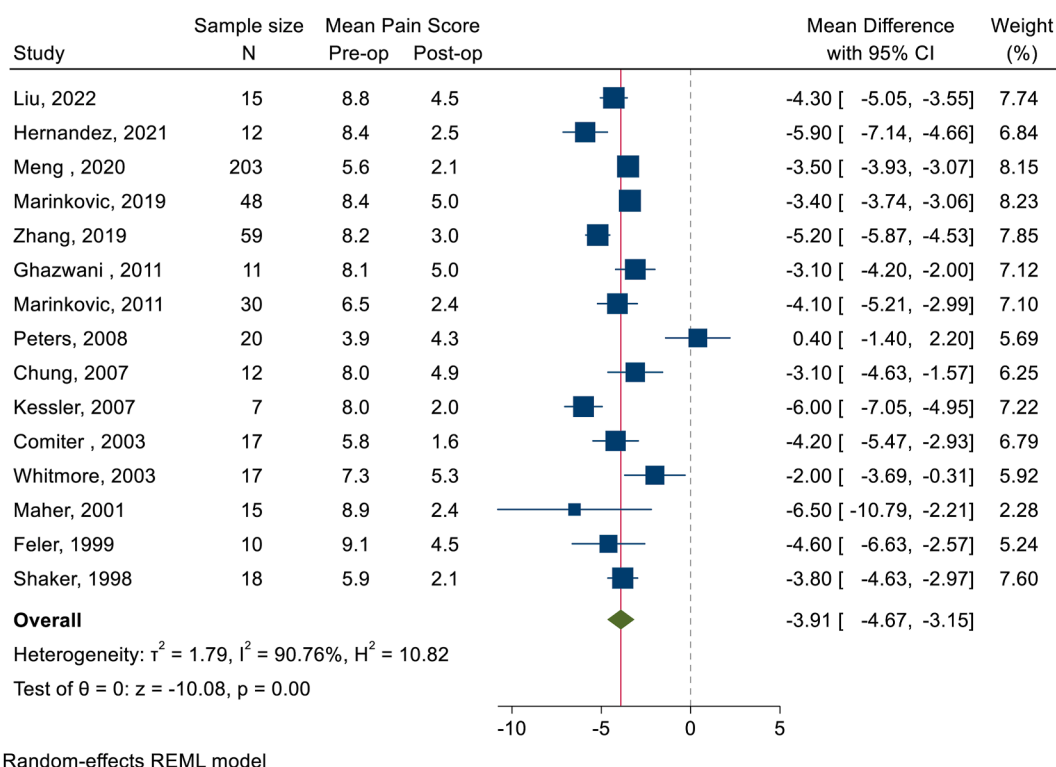


Figure 3. Pain score using mean difference. Post-op, postoperative; Pre-op, preoperative; REML, restricted maximum likelihood. [Color figure can be viewed at www.neuromodulationjournal.org]

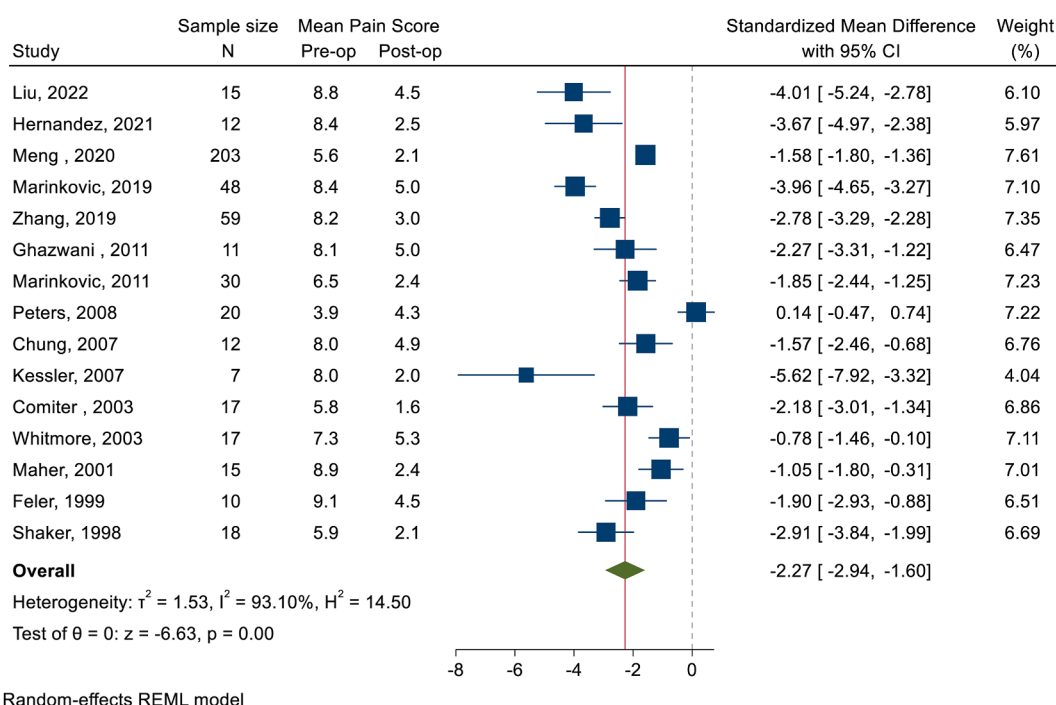


Figure 4. Pain score using SMD. Post-op, postoperative; Pre-op, preoperative; REML, restricted maximum likelihood. [Color figure can be viewed at www.neuromodulationjournal.org]

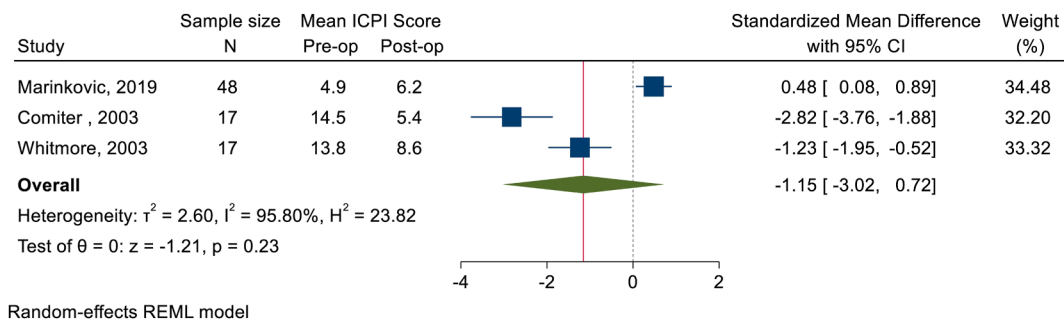


Figure 5. ICPI score using SMD. Post-op, postoperative; Pre-op, preoperative; REML, restricted maximum likelihood. [Color figure can be viewed at www.neuromodulationjournal.org]

- Increase in pain score: Peters et al¹⁵ reported an increase in NRS SMD of 0.14 (95% CI: -0.47, 0.74) at six-month follow-up.

ICSI and ICPI

The ICSI was fully reported in three studies with an overall SMD -2.18 (95% CI: -2.99, -1.37 $p < 0.005$) (Figs. 5 and 6).

- Best outcome: Marinkovic¹⁸ showed an SMD reduction in ICSI of -2.64 (95% CI: -3.18, -2.09).
- Limited data: Most studies did not report this metric, limiting cross-study comparisons.

ICPI also decreased in two of the three studies, with an SMD reduction of -1.15 (95% CI: -3.02, 0.72 $p = 0.23$). However, this was not statistically significant.

Quality of Life

QoL was a secondary outcome partially reported in ten of the 15 studies. QoL metrics varied but included validated tools such as the 36-Item Short Form Health Survey in addition to ad hoc scales. The meta-analysis results showed a statistically insignificant SMD worsening in QoL of 1.46 points (95% CI: 4.27, 1.35, $p = 0.31$) (Fig. 7).

- QoL improvements: Hernandez et al¹⁷ showed the only improvement in QoL score, with SMD of 2.67 (95% CI: 1.59, 3.75).
- Minimal changes: Meng et al¹⁴ indicated only a marginal QoL change (4.1–4.0 [95% CI: -0.26, 0.13]), reflecting the need for more robust QoL measures in future studies.

Voiding Frequency

The analysis found daily voiding frequency decreased significantly, with an SMD reduction of 1.71 episodes per 24 hours (95% CI: -2.29 to -1.12 $p < 0.05$) (Figs. 8 and 9).

- Significant improvement: Zhang et al¹⁹ showed a reduction from 25.0 to 15.7 voids per day (37.2% improvement).
- Least improvement: Chung et al²¹ reported an SMD reduction 1.8 voids per day, consistent with an insignificant efficacy in voiding control (95% CI: -8.93, 5.33).

Nocturia

Nocturia frequency, reported in nine studies, improved significantly post-SNS implantation with an SMD reduction of -1.49 (95% CI: -2.35, -0.63, $p < 0.001$). (Figs. 10 and 11).

- Largest reductions: Maher et al²⁵ reported an SMD reduction -4.70 episodes per night (95% CI: -6.08, 2.0).
- Increase in nocturia: Chung et al²¹ noted a mild increase in SMD nocturia of 0.33 (95% CI: -0.33, 0.99).

Complications and Explantation Rates

Complication rates varied greatly among studies, ranging from 0% to 40.6%. The mean complication rate was 16.3%. The most reported complications included lead migration, infection, and device malfunction. In addition, explantation rates ranged from 0% to 18%, with an average of 4.6%.

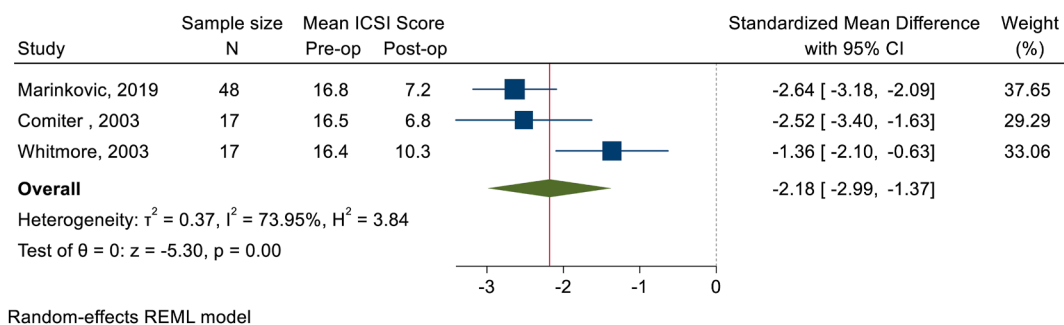


Figure 6. ICSI score using SMD. Post-op, postoperative; Pre-op, preoperative; REML, restricted maximum likelihood. [Color figure can be viewed at www.neuromodulationjournal.org]

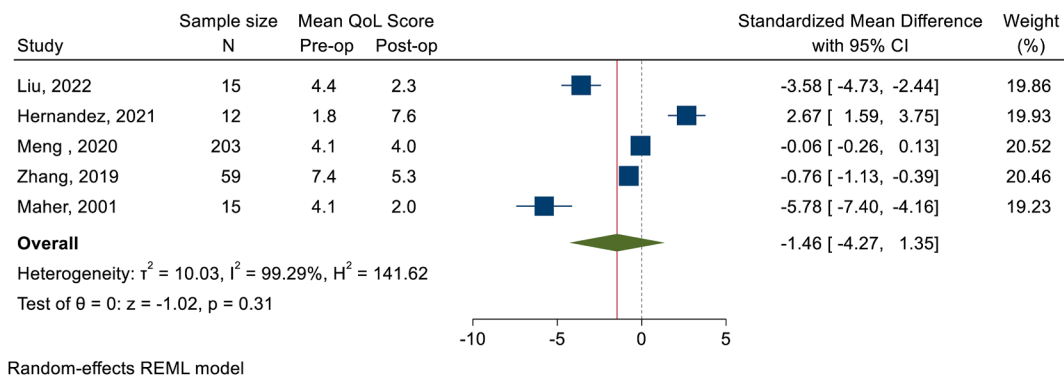


Figure 7. QoL score using SMD. Post-op, postoperative; Pre-op, preoperative; REML, restricted maximum likelihood. [Color figure can be viewed at www.neuromodulationjournal.org]

- High rates: Hernandez et al¹⁷ reported a 40.6% complication rate and a 9.4% explantation rate over a 75-month follow-up.
- Low rates: Liu et al¹⁶ and Comiter both²³ reported no complications or explantations over 12- and 14-month follow-ups, respectively.

DISCUSSION

Findings

The overall findings from the meta-analysis show that SNS is likely to have benefit in the management of CBPS. The included studies indicated statistically significant improvements in pain scores, ICSI, voiding, and nocturia symptoms but not in QoL or ICPI scores. Complication rates varied from 0% to 40%, as did explantation rates of 0% to 18%.

The meta-analysis revealed a generally high heterogeneity of studies, indicating very little standardization and the need for

larger sample sizes. The risk of bias overall was moderate. The studies included were generally of low-level evidence and quality, with no RCTs. Moreover, sample sizes were small, and powering was inadequate or not complete. The lack of standardization was most apparent in varied technique, outcome measures, and diagnosis/pathology. Specifically, CBPS was generally not identified as a separate entity under the umbrella term of CPP.

Previous Reviews and Key Differences

To date, the most comprehensive reviews and analysis were conducted by Wang et al in 2017¹² and Srivastava in 2012.²⁸ Since their reviews, several new studies were conducted, and five were added to this meta-analysis. In addition to the supplementary studies, there are several key differences between this study and the prior reviews.

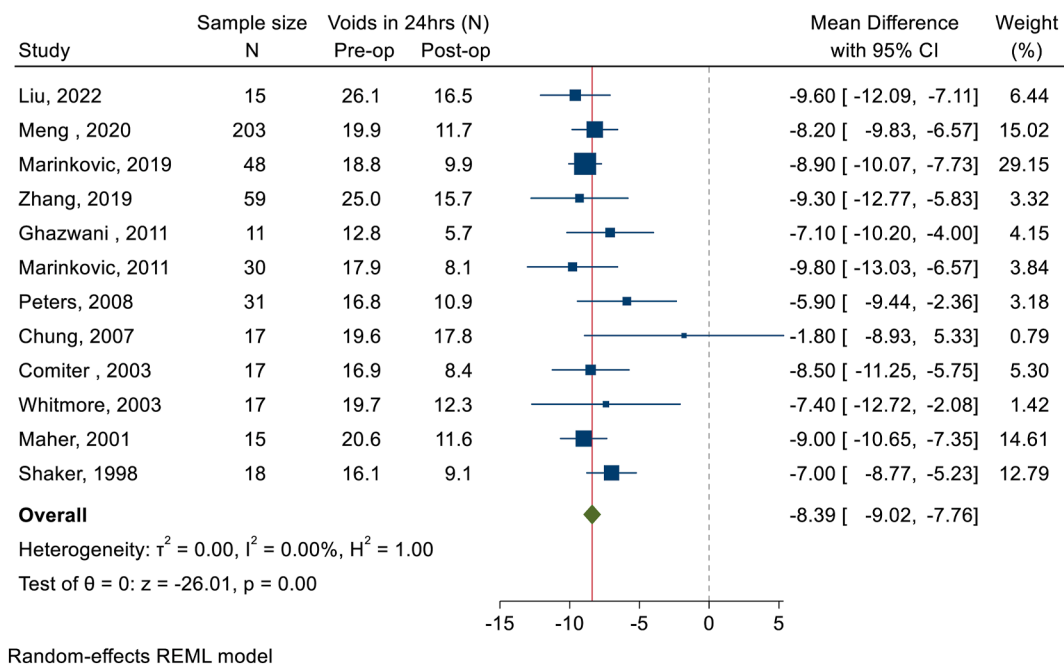


Figure 8. Twenty-four hour voids outcome using mean difference. Post-op, postoperative; Pre-op, preoperative; REML, restricted maximum likelihood. [Color figure can be viewed at www.neuromodulationjournal.org]

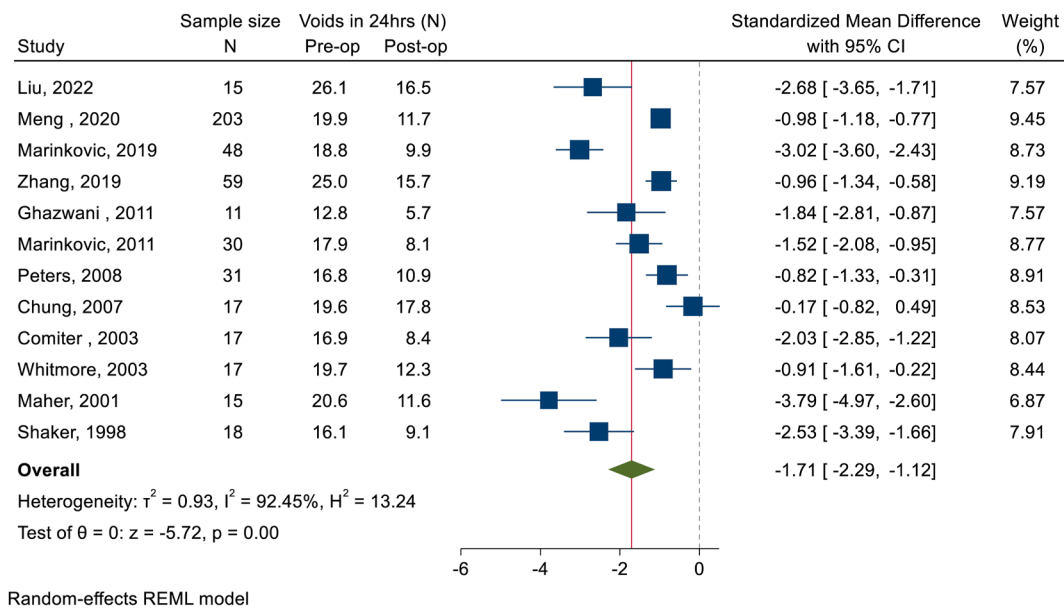


Figure 9. Twenty-four hour voids outcome using SMD. Post-op, postoperative; Pre-op, preoperative; REML, restricted maximum likelihood. [Color figure can be viewed at www.neuromodulationjournal.org]

The first key distinction lies in our inclusion criteria. Only studies that investigated patients with bladder pain and objectively measured pain scores were included; studies lacking specifically reported pain scores were excluded. This is important because most studies relied on heterogeneous diagnostic criteria, making it difficult to confirm whether the studied populations truly had CBPS. Consequently, the evaluation of SNS in these populations was often imprecise. This issue was further compounded by the interchangeable use of related terms such as IC and CPP. For instance, the study by Sokal et al,²⁹ included in the review by Wang et al, focused solely on patients with CPP—none of whom had symptoms consistent with CBPS. Some studies we excluded provided no objective outcome measure for pain but instead a

percentage change. This was evident in the study by Gajewski et al that provided Global Response Assessment percentage changes only, with no pre or post values.³⁰

The second major difference was that we excluded low-level evidence and incoherent methodological approaches. For example, the single-case report by Zermann et al was included in Srivastava's review,³¹ as was the study by Chai et al evaluating SNS through nerve testing rather than actual device implantation.³² Moreover, both previous reviews included the 2007 study by Peters et al, which compared peripheral nerve stimulation with SNS in patients with IC.³³ Although both reviews cited a 22-patient inclusion, an analysis of the original publication revealed that only four patients received SNS implants. This was somewhat confounded by

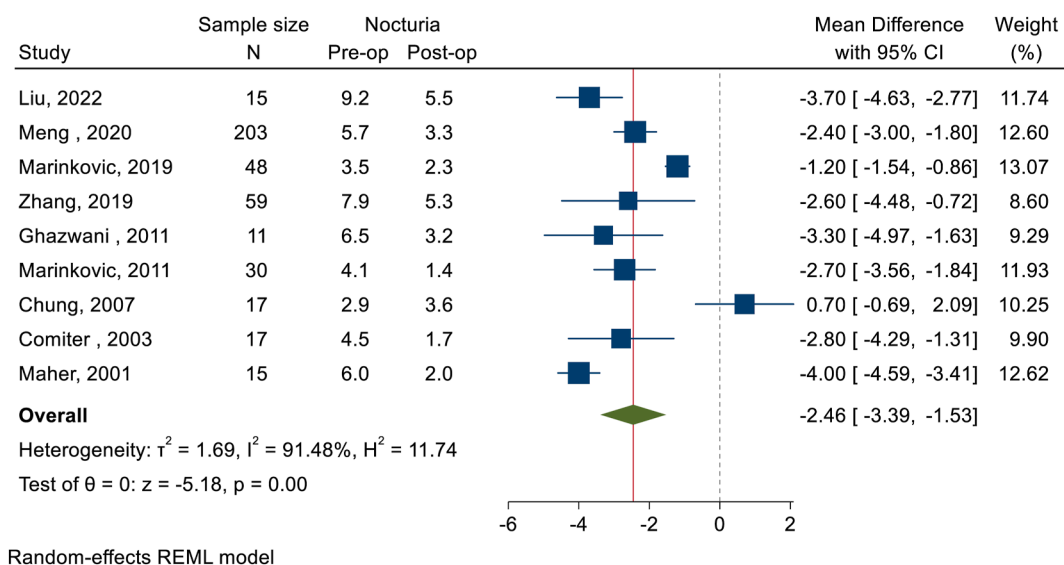


Figure 10. Nocturia outcome using mean difference. Post-op, postoperative; Pre-op, preoperative; REML, restricted maximum likelihood. [Color figure can be viewed at www.neuromodulationjournal.org]

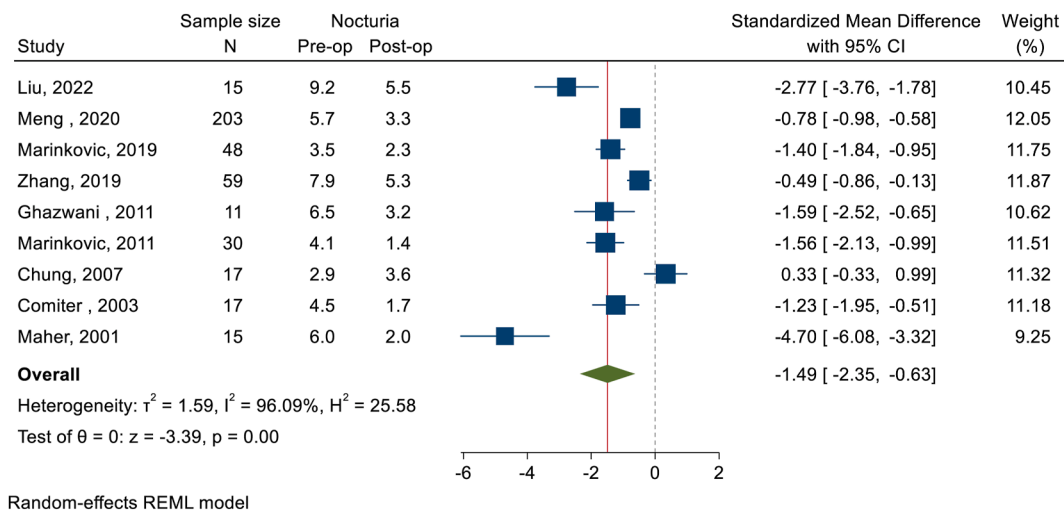


Figure 11. Nocturia outcome using SMD. Post-op, postoperative; Pre-op, preoperative; REML, restricted maximum likelihood. [Color figure can be viewed at www.neuromodulationjournal.org]

a further study by Peters et al in 2008 exploring the same topic.¹⁵ On the basis of the timelines and overlap, there is a strong possibility that the four patients from the RCT from 2007 also were included in the 2008 cohort. To avoid duplication, our review incorporated data only from the more recent prospective nonrandomized study.

Limitations

A key limitation of this meta-analysis is the substantial heterogeneity across all pooled analyses ($I^2 > 70\%$), indicating significant variability in study results. This likely stems from differences in study design, patient populations, interventions, and outcome measures. In general, most studies lacked key methodological details such as sample size calculations, statistical power considerations, and ways losses to follow-up were addressed. Although a random-effects model was used to account for between-study variability, high heterogeneity reduces the precision and generalizability of the pooled estimates. Moreover, many included studies had small sample sizes, which can limit statistical power and increase the risk of random error. Small studies also are more prone to inflated effect sizes and publication bias, particularly when only statistically significant results are reported. In addition, some of these studies were not designed to address CBPS specifically. These limitations warrant cautious interpretation of the findings and emphasize the need for future high-quality, well-powered studies to validate and refine these results.

Furthermore, a recent retrospective study on this topic was published after our search and analysis and thus could not be included. This was a further small retrospective study of SNS in CBPS with similar results: improved pain scores and alleviated urinary symptoms.³⁴ However, given the size and low level of evidence again, this would not significantly change our results or conclusions.

Opinion on the Role of SNS in CBPS

The overall results indicate that SNS may be helpful in the management of CBPS. However, it must be stressed that this remains only a possibility, given the level and quality of evidence were generally poor. As such, no confident inference can be made about the efficacy of SNS in CBPS.

Could SNS Be Effective for Functional Bladder Disorders and for Pain-Related Conditions?

Currently, SNS is used off label for CBPS, with its primary indication both in Europe and the USA being functional bladder issues. However, the overlap in function and pain is difficult to separate as delineated in the recent definitions update of CBPS in the EAU guidelines.³⁵ Our review highlights consistent discrepancies in diagnostic criteria and reporting adding to the confusion of the efficacy of SNS in CBPS.

Although a shared sacral nerve dysfunction is often suspected, the mechanisms behind functional and pain-based bladder syndromes remain poorly understood. Functional bladder disorders appear to have multifactorial causes—ranging from metabolic and hormonal imbalances to autonomic dysfunction and gut microbiota changes.³⁶ In contrast, CBPS is more closely linked to inflammatory processes, such as urothelial and neurogenic inflammation and vascular dysfunction.³⁷ Pain syndromes also align more with the biopsychosocial model, potentially affecting treatment strategies.³⁸

SNS is believed to modulate afferent and efferent pathways, restoring CNS–peripheral communication through inhibition, motor pathway stabilization, and central neuroplasticity.³⁹ Although effective in improving bladder and bowel function, its pain-relieving effects are less consistent, likely owing to the complexity of pain processing and the failure to acknowledge and manage the bio-psycho-social aspects. SNS may modulate spinal reflexes, reorganize cortical maps, and influence brain regions such as the periaqueductal gray.^{39,40} It also may engage mechanisms described in the gate control theory—blocking pain through activation of nonnociceptive fibers and descending inhibitory pathways.³⁹ Moreover, SNS can reduce excitatory neurotransmitters, enhance inhibitory neurotransmitters, and stimulate endogenous opioid release, collectively reducing pain perception.^{41–43}

What Does This Mean for the Future?

To draw definitive conclusions regarding the efficacy of SNS for CBPS, $\geq$ one well-designed, adequately powered RCT is essential. There also is the need to consider the effect of frequency of electrical stimulation, technique/anatomical approach, and the need for multidisciplinary team interventions. Although calling for such a

study is simple, conducting an RCT in the neuromodulation field poses practical challenges, including recruitment and funding.

Given the significant clinical and economic burden of CPP and CBPS, there is strong justification for institutional and provider support to explore effective treatment options such as SNS. SCS, a related modality, has revealed cost-effectiveness and improved QoL in conditions such as persistent spinal pain syndrome and complex regional pain syndrome.⁴⁴

A definitive trial in CBPS would not only provide clarity on the role of SNS in CBPS management but also guide better health care resource allocation and improve patient outcomes.

CONCLUSION

This meta-analysis highlights the potential of SNS as a therapeutic modality for CBPS. Although statistically significant improvements were observed in pain scores and urinary symptoms, the heterogeneity of study designs, inconsistent reporting, low-level evidence, and moderate-to-high risk of bias necessitate cautious interpretation. Future research should focus on adequately powered RCTs with standardized diagnostic criteria, reporting metrics, and follow-up durations to establish the clinical utility of SNS in CBPS management.

Declaration of AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used ChatGPT to help with planning, structure, and wording. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Authorship Statements

Rodwan Husein and Andrew Baranowski conducted the design, review, and search. Moein Tavakkoli helped prepare the protocol. Sara Ahmadi Abhari conducted the statistical analysis. Rodwan Husein prepared the manuscript with important intellectual input from Andrew Baranowski, Lawrence Hutchinson, Suraj Shah, and Anish Thillainathan. All authors approved the final manuscript.

Conflict of Interest

The authors reported no conflict of interest.

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SUPPLEMENTARY MATERIAL

To access the supplementary material accompanying this article, visit the online version of *Neuromodulation: Technology at the*

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COMMENTS

The Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) guidelines were met using the PRISMA checklist to ensure it. The information gathers all the information to date about SNS in BPS. Although there is high heterogeneity, the syndrome itself may vary among patients, which may be due to the lack of definite terminology; the article gives an insight into this treatment option. The authors explain the need for RCT in this pathology well.

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This systematic review and meta-analysis on SNS for CBPS is clearly structured and well written. Adherence to PRISMA guidelines and the focus on objective pain outcomes strengthen the work. The conclusions, however, are constrained by the low quality of the available evidence, characterized by the absence of RCTs, small sample sizes, heterogeneous diagnostic criteria, and a high risk of bias. Although the results are encouraging, they should be interpreted with caution. Well-powered, standardized RCTs are needed to confirm these findings.

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